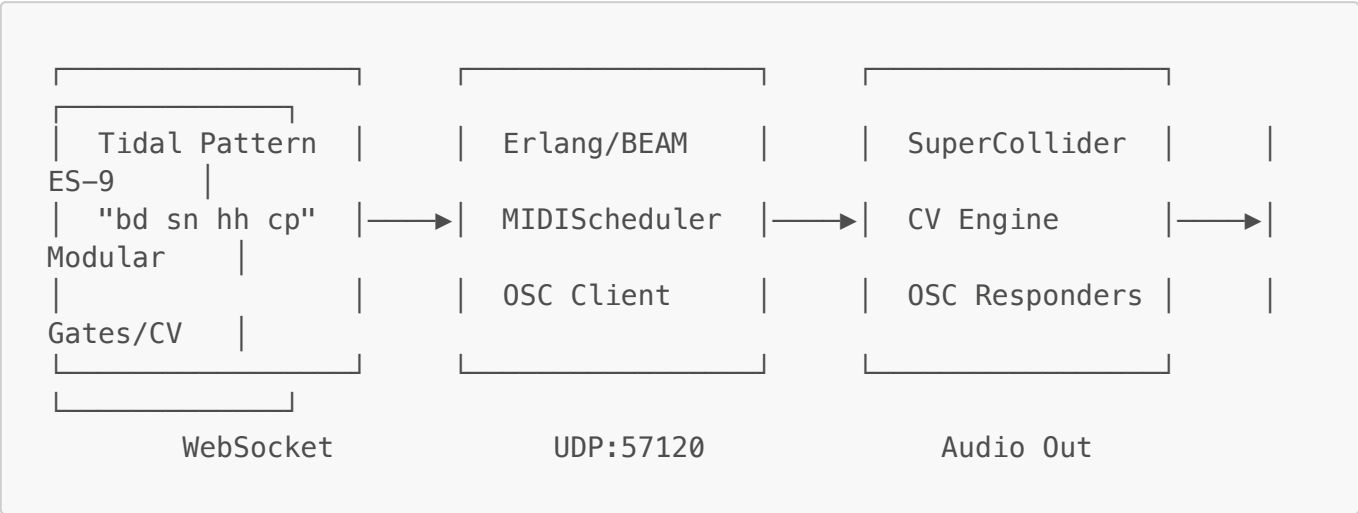


CV Output Setup for Expert Sleepers ES-9

This document captures all the nitty-gritty details for getting CV/Gate output from Tidal patterns to a modular synth via the Expert Sleepers ES-9.

Architecture



Critical Discoveries

1. ES-9 Channel Mapping

The ES-9's USB audio channels are NOT 1:1 with the panel jacks:

SuperCollider outbus	ES-9 Function
0-1	Headphone outputs
2-7	Unknown/unused
8	Jack 1 (first modular output)
9	Jack 2
10	Jack 3
...	...
15	Jack 8

Key insight: outbus: 8 = ES-9 jack 1. This offset varies by system - yours may differ!

2. SuperCollider SynthDef Fix

DC.ar() does NOT track control-rate changes. You MUST use K2A.ar():

```
// WRONG - won't respond to .set(\val, x)
SynthDef(\broken, { |out=0, val=0|
```

```

    Out.ar(out, DC.ar(val));
  }).add;

// CORRECT - responds to .set(\val, x)
SynthDef(\working, { |out=0, val=0|
  Out.ar(out, K2A.ar(val));
}).add;

```

3. Don't Overlap CV and Gate Synths

If you create both a CV synth and a Gate synth on the same output bus, they interfere. The CV engine splits them:

- **Gates:** Channels 0-3 → ES-9 jacks 1-4 (outbus 8-11)
- **CV:** Channels 0-3 → ES-9 jacks 5-8 (outbus 12-15)

4. Erlang Build Requires `erlc`

The Makefile now includes this, but if builds seem stale:

```

cd showcases/psd3-tilted-radio/purerl-tidal
find output -name "*.erl" -exec erlc -o ebin {} \;

```

5. rebar3 Shell Code Path

The purerl-compiled beam files are in `ebin/` but rebar3 shell doesn't automatically include it:

```

code:add_path("/Users/afc/work/afc-work/PSD3-Repos/showcases/psd3-tilted-
radio/purerl-tidal/ebin").

```

Complete Startup Procedure

Step 1: Build

```

cd /Users/afc/work/afc-work/PSD3-Repos
make app-tilted-radio

```

Step 2: Start SuperCollider

Open SuperCollider and run (BEFORE booting server):

```

// Configure for ES-9 - run this BEFORE s.boot
s.options.outDevice = "ES-9";

```

```
s.options.inDevice = "ES-9";
s.options.numOutputBusChannels = 16;
s.options.numInputBusChannels = 16;
s.options.sampleRate = 48000;
s.boot;
```

Verify:

```
s.options.outDevice;           // "ES-9"
s.options.numOutputBusChannels; // 16
```

Step 3: Load CV Engine

Open and evaluate (Cmd+Enter on entire file):

```
/Users/afc/work/afc-work/PSD3-Repos/showcases/psd3-tilted-radio/purerl-
tidal/supercollider/tidal-cv-engine.scd
```

You should see:

```
Persistent synths created:
  Gate channels 0-3 → ES-9 jacks 1-4 (outbus 8-11)
  CV channels 0-3 → ES-9 jacks 5-8 (outbus 12-15)
=== Tidal CV Engine Ready ===
```

Test it works:

```
~testGate.(0, 1); // Jack 1 LED should light
~testGate.(0, 0); // Jack 1 LED should turn off
~testCV.(0, 0.5); // Jack 5 should show half voltage
```

Step 4: Start Erlang Server

```
# Kill any existing process on port 8080
lsof -ti:8080 | xargs kill -9 2>/dev/null

# Start rebar3 shell
cd /Users/afc/work/afc-work/PSD3-Repos/showcases/psd3-tilted-radio/purerl-
tidal
rebar3 shell
```

In Erlang shell:

```
code:add_path("/Users/afc/work/afc-work/PSD3-Repos/showcases/psd3-tilted-
radio/purerl-tidal/ebin").
('main@ps':main())().
```

You should see:

```
MIDI Scheduler started
Device: IAC Driver Tidal
BPM: 120
Pattern: ~
Gate output: enabled (OSC 127.0.0.1:57120)
```

Step 5: Send Patterns

```
wscat -c ws://localhost:8080/ws
```

Then type patterns:

```
bd sn hh cp
bd*4
bd(3,8)
[bd sn] hh*2
```

Watch ES-9 jack 1 flash with each drum hit!

Troubleshooting

No output on ES-9 jacks

1. Check SuperCollider is using ES-9: `s.options.outDevice`
2. Check channel count: `s.options.numOutputBusChannels` (should be 16)
3. Test direct output: `{ DC.ar(0.5) }.play(outbus: 8);`
4. If that works but CV engine doesn't, reload the .scd file

Gate output not appearing in Erlang console

1. Check for "Gate output: enabled" at startup
2. If missing, rebuild: `make app-tilted-radio`
3. Check beam files are fresh: `ls -la ebin/main@ps.beam`

"undefined function" in Erlang

Add the code path:

```
code:add_path("/Users/afc/work/afc-work/PSD3-Repos/showcases/psd3-tilted-radio/purerl-tilal/ebin").
```

Port 8080 already in use

```
lsof -ti:8080 | xargs kill -9
```

File Locations

File	Purpose
supercollider/tilal-cv-engine.scd	SuperCollider CV/Gate engine
src/Tidal/OSC.purs	PureScript OSC bindings
src/Tidal/OSC.erl	Erlang OSC FFI (sendCV, sendGate, etc.)
src/Tidal/MIDIScheduler.purs	Main scheduler with gate integration
src/Main.purs	Entry point with gate config

Configuration

In `src/Main.purs`, the gate config:

```
gateConfig = defaultGateConfig
{ enabled = true
, oscHost = "127.0.0.1"
, oscPort = 57120      -- sclang default
, channelOffset = 0    -- ch10 -> gate 0
, gateDuration = 50.0  -- 50ms pulse
}
```

In `supercollider/tilal-cv-engine.scd`:

```
~es90offset = 8;           // outbus 8 = ES-9 jack 1
~gateBusOffset = 8;        // Gates on jacks 1-4
~cvBusOffset = 12;         // CVs on jacks 5-8
~cvScale = 0.5;            // Max +5V (conservative)
```

Next Steps

- ☐ CV patterns for pitch control (Pattern Number)
 - ☐ Multiple tracks on different gate channels
 - ☐ ES-5 ADAT outputs for more gates
 - ☐ ESX-8GT integration for dedicated gate outputs
-

Last updated: 2026-01-12 Tested with: SuperCollider 3.x, Erlang/OTP 28, ES-9 firmware latest